GAAFET Numerical Formula Chart

3-Stack Silicon Nanosheet NMOS - Experiment, Extraction, PPA and ML

Prepared for the current Phase-1 experiment: L_g, W_{NS} and T_{NS} DOE

A. Phase-1 variables - vary only these

Parameter	Symbol	Range
Gate length	L_g	10–20 nm
Nanosheet width	W_{NS}	15–30 nm
Sheet thickness	T_{NS}	4–8 nm

Recommended first full-factorial grid: 6 levels per factor.

$$N_{DOE} = N_{L_g} N_W N_T = 6 \times 6 \times 6 = 216$$

This directly matches the supervisor recommendation of roughly 200–250 TCAD geometries with proper DOE.

B. Freeze for Phase 1

$N_{sheet} = 3$, Si channel, sheet spacing, dielectric stack, metal gate, work function, S/D structure and doping, substrate doping, BDI, temperature and physics models.

Reason: changes in the output must be attributable to the 3 DOE geometry variables, not to architecture changes.

C. Baseline values already obtained

SS	65.0 mV/dec
$DIBL$	10.32 mV/V
$V_{th,sat}$	0.070 V
I_{ON}	1.438 mA/μm
I_{OFF}	0.216 pA/μm
$\log_{10}(I_{ON}/I_{OFF})$	9.82

Use these as the **reference row** when testing every new extraction script.

D. Bias points - keep identical across DOE

$I_D - V_G$ **low-V_D:** $V_{DS} = 0.05$ V, $V_{GS} : 0 \rightarrow 0.70$ V.
$I_D - V_G$ **high-V_D:** $V_{DS} = 0.70$ V, $V_{GS} : 0 \rightarrow 0.70$ V.
$I_D - V_D$**:** $V_{DS} : 0 \rightarrow 0.70$ V at $V_{GS} = 0, 0.2, 0.4, 0.6, 0.7$ V.

Do not change extraction definitions midway through the dataset.

E. Minimum dataset row

Inputs:

$$X = [L_g, W_{NS}, T_{NS}]$$

Primary TCAD outputs:

$$Y = [V_{th}, SS, DIBL, I_{ON}, I_{OFF}, g_m, g_{ds}, C_{gg}]$$

Derived PPA outputs:

$$[P_{leak}, \tau_{int}, I_{ON}/I_{OFF}]$$

Optional advanced outputs: R_{ON}, r_o, f_T, C_{gs}, C_{gd}, PDP.

F. One rule that prevents most numerical errors

Always write the **bias point + normalization basis + unit** beside every value. Example:
$I_{ON} = I_D(V_G = 0.7, V_D = 0.7)$ in mA/μm, where μm is the chosen effective-width normalization.
A number without its extraction condition is not a reusable dataset value.

1. Electrical Parameter Extraction from $I_D - V_G$ and $I_D - V_D$

1. Threshold voltage - maximum-g_m method

$$g_m = \frac{\partial I_D}{\partial V_G}$$

Find the gate voltage $V_{G,m}$ where g_m is maximum. The tangent-intercept estimate is

$$V_{th} \approx V_{G,m} - \frac{I_D(V_{G,m})}{g_{m,max}}$$

Use the same method for every geometry. Report separately as $V_{th,lin}$ at low V_D and $V_{th,sat}$ at high V_D.

2. Subthreshold swing (SS)

Between two points in the straight subthreshold region of the semi-log $I_D - V_G$ curve:

$$SS = \frac{\Delta V_G}{\Delta \log_{10}(I_D)} \times 1000 \quad [\text{mV/dec}]$$

Derivative form:

$$SS = \left(\frac{d \log_{10} I_D}{dV_G}\right)^{-1} \times 1000$$

At 300 K, the ideal thermal reference is

$$SS_{ideal} = \ln(10)\frac{kT}{q} \approx 59.6 \text{ mV/dec}$$

A useful electrostatic sanity relation is

$$SS \approx SS_{ideal}\left(1 + \frac{C_{dep} + C_{it}}{C_{ox}}\right)$$

3. DIBL

Using the same V_{th} extraction method at low and high drain bias:

$$DIBL = \frac{V_{th,lowD} - V_{th,highD}}{V_{D,high} - V_{D,low}} \times 1000 \quad [\text{mV/V}]$$

For this project, $V_{D,low} = 0.05$ V and $V_{D,high} = 0.70$ V.

4. ON current, OFF current and ON/OFF ratio

Recommended project definitions:

$$I_{OFF} = I_D(V_G = 0,\ V_D = V_{DD})$$

$$I_{ON} = I_D(V_G = V_{DD},\ V_D = V_{DD})$$

$$R_{ON/OFF} = \frac{I_{ON}}{I_{OFF}}$$

For large ratios, store the log value too:

$$R_{log} = \log_{10}\left(\frac{I_{ON}}{I_{OFF}}\right)$$

For ML, predicting $\log_{10}(I_{OFF})$ is often numerically better than predicting raw leakage spanning many decades.

5. Transconductance and output conductance

$$g_m = \left.\frac{\partial I_D}{\partial V_G}\right|_{V_D} \qquad g_{ds} = \left.\frac{\partial I_D}{\partial V_D}\right|_{V_G}$$

Discrete central-difference estimate:

$$g_m(V_i) \approx \frac{I_D(V_{i+1}) - I_D(V_{i-1})}{V_{G,i+1} - V_{G,i-1}}$$

Likewise for g_{ds} versus V_D.

If I_D is in mA/μm and voltage in V, g_m and g_{ds} are in mS/μm.

6. Resistance from output curves

Small-signal output resistance in saturation:

$$r_o = \frac{1}{g_{ds}}$$

Low-V_D effective ON resistance at a stated gate bias:

$$R_{ON} \approx \frac{V_{DS}}{I_D} \quad \text{for small } V_{DS}$$

Never use a high-V_D saturation point to label R_{ON} without stating the convention.

2. Geometry, Effective Width, Capacitance and High-k Checks

7. Effective gate-wrapped width for rectangular nanosheets

For one rectangular nanosheet fully wrapped by gate, the gated perimeter is approximately

$$W_{eff,1} = 2(W_{NS} + T_{NS})$$

For N_{sheet} identical sheets:

$$W_{eff,total} = N_{sheet}\, 2(W_{NS} + T_{NS})$$

Current normalization:

$$I_D^{norm} = \frac{I_D^{raw}}{W_{eff,total}}$$

Baseline geometry: $N = 3$, $W_{NS} = 15$ nm, $T_{NS} = 4$ nm gives

$$W_{eff,total} = 3 \times 2(15 + 4) = 114 \text{ nm} = 0.114\ \mu m$$

Use this only if your project chooses gate-perimeter normalization. Do not mix this with physical sheet width normalization.

8. Channel and gated surface geometry

Total silicon cross-sectional area of the three sheets:

$$A_{Si,total} = N_{sheet} W_{NS} T_{NS}$$

Approximate gated surface area used for oxide-capacitance sanity checks:

$$A_{gate} \approx W_{eff,total} L_g$$

These are **not** the same as layout footprint area.

Area / footprint warning

The supervisor specifically asked what "Area" means. Do not report $L_g W_{NS}$ as device PPA area unless a layout footprint convention is formally defined. For Phase 1, store geometry dimensions and call any constructed area a **geometric proxy**, not fabricated footprint.

9. Oxide capacitance density

$$C'_{ox} = \frac{\varepsilon_0 \kappa_{ox}}{t_{ox}} \quad [\mathrm{F/m^2}]$$

Convenient units:

$$C'_{ox}[\mathrm{fF}/\mu\mathrm{m}^2] \approx 8.854 \frac{\kappa_{ox}}{t_{ox}[\mathrm{nm}]}$$

Approximate oxide capacitance over the gated area:

$$C_{ox} \approx C'_{ox} A_{gate}$$

10. EOT for a dielectric stack

For layers i with thickness t_i and relative permittivity κ_i:

$$EOT = \sum_i t_i \frac{\kappa_{SiO_2}}{\kappa_i}, \qquad \kappa_{SiO_2} \approx 3.9$$

For a physical SiO_2 interfacial layer plus HfO_2:

$$EOT = t_{SiO_2} + t_{HfO_2} \frac{3.9}{\kappa_{HfO_2}}$$

Use this later when high-k thickness becomes a Phase-2 parameter.

11. Gate capacitance from TCAD charge

The quantity needed for device-level delay is C_{gg}, not automatically C_{ox}:

$$C_{gg} = \frac{\partial Q_G}{\partial V_G}$$

If terminal components are available:

$$C_{gg} \approx C_{gs} + C_{gd} + C_{gb}$$

Important: C_{gg} includes device electrostatics/quantum response; C_{ox} is an oxide-stack quantity. They should not be used interchangeably.

3. Device-Level Power, Performance and PPA Formulas

12. Leakage / static power - from supervisor comments

$$P_{static} = P_{leak} = I_{OFF}V_{DD}$$

Useful shortcut for normalized data:

$$P_{leak}[\mathrm{pW}/\mu\mathrm{m}] = I_{OFF}[\mathrm{pA}/\mu\mathrm{m}] \times V_{DD}[\mathrm{V}]$$

Baseline:

$$P_{leak} = 0.216 \times 0.7 = 0.1512\ \mathrm{pW}/\mu\mathrm{m}$$

Use the term **leakage power** or **static power**, as recommended in the project comments.

13. Dynamic power - from supervisor comments

$$P_{dynamic} = \alpha C_{eff} V_{DD}^2 f$$

$$C_{eff} = C_g + C_{diff} + C_{interconnect}$$

Convenient unit shortcut:

$$P_{dynamic}[\mu W] = \alpha\, C_{eff}[\mathrm{fF}]\, V_{DD}^2\, f[\mathrm{GHz}]$$

Device-level caution: a single TCAD transistor does not inherently contain the final interconnect capacitance. Do not claim full circuit dynamic power unless $C_{interconnect}$ is defined externally.

14. Switching energy

A useful companion quantity is

$$E_{switch} = \frac{1}{2} C_{eff} V_{DD}^2$$

Use only after the capacitance convention and switching event are stated.

15. Intrinsic delay - recommended project metric

For device-level performance:

$$\tau_{int} = \frac{C_{gg}V_{DD}}{I_{ON}}$$

Very convenient normalized-unit form:

$$\tau_{int}[\mathrm{ps}] = \frac{C_{gg}[\mathrm{fF}/\mu\mathrm{m}]\, V_{DD}[\mathrm{V}]}{I_{ON}[\mathrm{mA}/\mu\mathrm{m}]}$$

Lower τ_{int} means better device-level speed.

16. Small-signal speed metrics - optional

A second characteristic time is

$$\tau_{gm} = \frac{C_{gg}}{g_m}$$

Approximate unity-current-gain frequency:

$$f_T \approx \frac{g_m}{2\pi C_{gg}}$$

Unit shortcut:

$$f_T[\mathrm{GHz}] \approx 159.15 \frac{g_m[\mathrm{mS}]}{C_{gg}[\mathrm{fF}]}$$

Use f_T only when g_m and C_{gg} are extracted at the **same bias point**.

17. Power-delay combinations - optional ranking

$$PDP = P\,\tau \qquad EDP = P\,\tau^2$$

Use these only as secondary figures of merit. For the paper, keep the original physical metrics visible so the optimization trade-off remains interpretable.

PPA naming discipline

For this project, a defensible Phase-1 wording is **device-level PPA**: leakage/static power + intrinsic delay/performance + explicitly defined geometry/footprint proxy. Do not silently equate transistor geometry with standard-cell area.

4. Literature Comparison, DOE, ML Accuracy and Optimization

18. Literature / benchmark numerical comparison

Absolute deviation:

$$\Delta x = x_{TCAD} - x_{ref}$$

Absolute error magnitude:

$$|\Delta x| = |x_{TCAD} - x_{ref}|$$

Relative deviation:

$$\Delta_{\%} = \frac{x_{TCAD} - x_{ref}}{x_{ref}} \times 100\%$$

Absolute percentage error:

$$APE = \left|\frac{x_{TCAD} - x_{ref}}{x_{ref}}\right| \times 100\%$$

For leakage spanning decades, compare logarithms:

$$\Delta_{dec} = \log_{10}(I_{OFF,TCAD}) - \log_{10}(I_{OFF,ref})$$

19. Full-factorial DOE count

$$N_{DOE} = \prod_{j=1}^{p} N_j$$

For the current 3-factor, 6-level design:

$$N_{DOE} = 6^3 = 216$$

Recommended levels used in the work plan:
$L_g = 10, 12, 14, 16, 18, 20$ nm
$W_{NS} = 15, 18, 21, 24, 27, 30$ nm
$T_{NS} = 4, 4.8, 5.6, 6.4, 7.2, 8$ nm

Train/test rule from supervisor comments

Do not rely only on a random split. Leave out one or more complete geometries / geometry levels and test whether the model predicts unseen L_g, W_{NS} or T_{NS} combinations.

20. ML regression metrics

For measured/TCAD targets y_i and ML predictions $\hat{y}_i$:

$$MAE = \frac{1}{N}\sum_{i=1}^{N} |y_i - \hat{y}_i|$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

Percentage metric when the target is safely away from zero:

$$MAPE = \frac{100}{N}\sum_i \left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

For I_{OFF}: prefer MAE/RMSE on $\log_{10}(I_{OFF})$ because MAPE can be unstable for extremely small leakage.

21. Normalization before multi-objective optimization

Min-max scaling:

$$z = \frac{x - x_{min}}{x_{max} - x_{min}}$$

For a metric to be maximized, one minimization-form option is

$$f_{max} = 1 - z$$

Typical NSGA-II objective directions:
minimize P_{leak}, τ_{int}, SS, $DIBL$; maximize I_{ON} or g_m subject to constraints on V_{th} and I_{OFF}.

22. Pareto dominance - interpretation

Solution A dominates B when A is no worse than B in every objective and strictly better in at least one.
Practical rule: report a Pareto front, not a single "best" GAAFET, unless application-specific weights are declared.

5. Worked Baseline Checks + Numerical Worksheet

23. Baseline ON/OFF ratio check

Given $I_{ON} = 1.438$ mA/μm and $I_{OFF} = 0.216$ pA/μm:

$$\frac{I_{ON}}{I_{OFF}} = \frac{1.438 \times 10^{-3}}{0.216 \times 10^{-12}} = 6.66 \times 10^9$$

$$\log_{10}(I_{ON}/I_{OFF}) \approx 9.82$$

This reproduces the benchmark report value and is a useful spreadsheet sanity check.

24. Baseline DIBL threshold-shift check

For $DIBL = 10.32$ mV/V and $\Delta V_D = 0.70 - 0.05 = 0.65$ V:

$$\Delta V_{th} = 10.32 \times 0.65 = 6.708 \text{ mV}$$

If $V_{th,sat} = 70.0$ mV, then the corresponding low-V_D threshold implied by this DIBL is approximately

$$V_{th,lin} \approx 70.0 + 6.708 = 76.708 \text{ mV}$$

Use this only as a consistency check; final values must come from the actual extracted curves.

25. Baseline SS body-factor sanity check

At 300 K:

$$m \equiv \frac{SS}{SS_{ideal}} \approx \frac{65.0}{59.6} = 1.091$$

A value close to 1 is consistent with strong electrostatic gate control. This is a sanity indicator, not a substitute for full electrostatic calibration.

26. Fill one row for every TCAD geometry

L_g **(nm)**	______	W_{NS} **(nm)**	______
T_{NS} **(nm)**	______	W_{eff} **(μm)**	______
$V_{th,lin}$ **(V)**	______	$V_{th,sat}$ **(V)**	______
SS (mV/dec)	______	**DIBL (mV/V)**	______
I_{ON} **(mA/μm)**	______	I_{OFF} **(pA/μm)**	______
$g_{m,max}$ **(mS/μm)**	______	g_{ds} **(mS/μm)**	______
C_{gg} **(fF/μm)**	______	P_{leak} **(pW/μm)**	______
τ_{int} **(ps)**	______	$\log_{10}(I_{ON}/I_{OFF})$	______

27. Numerical consistency checklist

- Same V_D values for all DIBL extractions.
- Same threshold method for all geometries.
- Same current-normalization convention for TCAD and literature comparison.
- Same temperature and physics models across Phase 1.
- Keep raw current and normalized current in separate columns.
- Keep raw TCAD outputs separate from derived formulas.
- Store logarithmic leakage columns in addition to raw leakage.
- Re-run 3–4 baseline cases before launching all 216 geometries.

28. Recommended spreadsheet column order

`RunID | Lg | Wns | Tns | Weff | Vth_lin | Vth_sat | SS | DIBL | Ion | Ioff | logIoff | gm_max | gds | Cgg | Pleak | tau_int | logIonIoff | convergence_flag`

Basis: current GAAFET benchmark report and supervisor/project comments. Source-derived items include recommended ranges, 200–250 DOE target, power equations, requested delay/g_m/area/C_G clarification, Random Forest/XGBoost starting models, held-out-geometry testing, NSGA-II and physics-informed interpretation. Other formulas are recommended operational definitions for this TCAD workflow and should be kept fixed once the experiment begins.